

Facial-Expression-Aware Prompting for Empathetic LLM Tutoring

Shuangquan Feng¹ Laura Fleig² Ruisen Tu³ Philip Chi³ Edmund Bu^{4,5}
 Melinda Ozel⁶ Junhua Ma³ Teng Fei² Virginia R. de Sa^{2,7}

¹ Neurosciences Graduate Program, University of California San Diego, La Jolla, USA

² Department of Cognitive Science, University of California San Diego, La Jolla, USA

³ Department of Computer Science and Engineering, University of California San Diego, La Jolla, USA

⁴ Department of Electrical and Computer Engineering, University of California San Diego, La Jolla, USA

⁵ Department of Mathematics, University of California San Diego, La Jolla, USA

⁶ Face the FACS

⁷ Halicioğlu Data Science Institute, University of California San Diego, La Jolla, USA

Abstract

Large language models (LLMs) enable increasingly capable tutoring-style conversational agents, yet effective tutoring requires sensitivity to learners’ affective and cognitive states beyond text alone. Facial expressions provide immediate and practical cues of confusion, frustration, or engagement, but remain underexplored in LLM-driven tutoring. We investigate whether facial-expression-aware signals can improve empathetic tutoring responses through prompt-level integration, without end-to-end retraining. We build a scalable simulated tutoring environment where a student agent exhibits diverse facial behaviors from a large unlabeled human facial expression video dataset, and compare four tutor variants: a text-only LLM baseline, a multimodal baseline using a random facial frame, and two Action Unit estimation model (AUM)-based methods that either inject textual AU descriptions or select a peak-expression frame for visual grounding. Across 960 multi-turn conversations spanning three tutor backbones (GPT-5.1, Claude Opus 4.5, and Gemini 2.5 Pro), we evaluate targeted pairwise comparisons with five human raters and an exhaustive AI evaluator. AU-based conditioning consistently improves empathetic responsiveness to facial expressions across all tutor backbones, while AUM-guided peak-frame selection outperforms random-frame visual input. Textual AU abstraction and peak-frame visual injection show model-dependent advantages. Control analyses show that this improvement does not come at the expense of worse pedagogical clarity or responsiveness to textual cues. Finally, AI-human agreement is highest on

facial-expression-grounded empathy, supporting scalable AI evaluation for this dimension. Overall, our results show that lightweight, structured facial expression representations can meaningfully enhance empathy in LLM-based tutoring systems with minimal overhead. Future work should evaluate these methods using real student facial data collected during authentic learning interactions, and examine whether improved empathetic responsiveness translates into measurable benefits in engagement and learning outcomes.

1 Introduction

Large language models (LLMs) have enabled increasingly capable conversational systems in domains such as customer support and education [52, 54, 56, 59]. However, effective human-aligned interaction often requires sensitivity to users’ affective and cognitive states, especially when users provide limited textual feedback [46, 55]. Multimodal signals can provide complementary cues for such adaptation [2, 10, 22], and facial expressions are particularly practical indicators of states like confusion, frustration, or engagement [18]. Despite this, facial behavior remains underexplored in LLM-driven interaction frameworks compared to text or audio [20].

AI tutoring presents a particularly compelling and intuitive scenario to demonstrate the feasibility and value of incorporating empathy into multimodal LLMs (MLLMs). Effective tutoring depends in part on the tutor’s ability to recognize and respond to learners’ emotional and cognitive states, such as confusion and engagement. Recent work on LLM-based tutoring systems has shown their potential to provide personalized, scalable, and effective educational support [8, 47, 54]. In this context, AI tutoring provides a

*We are grateful for support from the California Education Learning Lab, seed funding from UC San Diego Social Sciences and the Sanford Institute for Empathy and Compassion, funds from the Halicioğlu Data Science Institute Endowed Chair I, and anonymous donor funds to Cognitive Science as well as hardware funding from NVIDIA, Adobe, and Sony.

clear and practical domain for examining the feasibility and value of empathetic multimodal interaction. In this work, we investigate whether injecting facial-expression-aware signals into tutoring-style LLM responses can improve empathetic responsiveness without end-to-end retraining. We incorporate structured facial representations derived from Action Units (AUs) and compare two lightweight integration strategies: (i) converting AU estimates into concise textual descriptions prepended to a text-only tutor prompt, and (ii) using AU-based saliency to select a peak frame for a multimodal tutor. Through controlled simulated multi-turn tutoring conversations and both human and AI evaluation, we analyze when and how facial-expression-awareness improves empathy in tutoring responses.

2 Related Work

2.1 Affective Intelligence in Education

AI systems increasingly support education through intelligent tutors, automated grading tools, and adaptive learning platforms. However, many systems primarily focus on cognitive tasks (e.g. content delivery, problem-solving) while overlooking emotional states that significantly influence learning [58]. Affective intelligence (the ability of a system to detect, interpret, and respond to human emotions) has thus become increasingly important in the design of educational AI. In tutoring systems, recognizing affective cues such as confusion or frustration allows for better-timed and more effective support, enhancing engagement and learning [11].

2.2 Facial Expression Recognition

To effectively detect emotional cues, researchers have explored a variety of multimodal signals. Among them, facial expressions are particularly rich in affective content and often provide the clearest window into emotional states. In educational settings, facial expressions serve as crucial indicators of student engagement and comprehension [5, 44, 50].

Traditional facial expression analysis outputs one of two types of responses to the images/videos [28]: (1) associated emotional/cognitive states or levels of valence/arousal, which are used in popular datasets like AffectNet [41], FERPlus [4, 17], RAF-DB [29], Aff-Wild [62], and Aff-Wild2 [26]; (2) facial action unit (AU) activation, as defined in the Facial Action Coding System (FACS, a comprehensive system breaking down facial expressions into individual components of AUs of muscle movement [12]). AUs offer anatomically grounded, composable descriptors of facial movement, enabling nuanced emotion inference, especially in domains like education where emotional states are difficult to estimate [3] and often fall outside of traditional

emotional taxonomies [57]. Deep learning-based AU detectors trained on datasets, including BP4D [64], DISFA and DISFA+ [38–40], EmotionNet [13], and others, have advanced FER capabilities [7, 15, 21, 24, 25, 30, 31, 37, 45, 48, 49, 51, 60, 61, 65–67].

2.3 MLLMs and Visual Grounding of Affective Cues

A prerequisite for our proposed approach for empathetic LLM tutoring is that LLMs and MLLMs can meaningfully interpret facial affective cues. Recent evidence supports this. [33] benchmarked GPT-4V on generalized emotion recognition across 21 datasets, finding that it outperforms supervised systems on some visual sentiment tasks and demonstrates non-trivial facial emotion recognition ability. Emotion-specialized models push this further: Emotion-LLaMA [6] integrates emotion-specific visual encoders with instruction tuning to improve fine-grained multimodal emotion understanding, while AU-LLaVA [19] and AULLM++ [35] demonstrate that LLMs can perform AU detection when visual features are injected as textual semantic premises. A complementary prompting paradigm converts non-textual affective signals, such as AU descriptions, into tagged text snippets for LLM consumption [32], and [63] show that visual prompting strategies can guide vision-language models to enhance emotion recognition. Collectively, these works establish that modern LLMs possess a usable capacity to decode facial expressions, whether through native visual processing or textual abstraction of facial cues. However, this capacity has been explored exclusively for emotion recognition, a classification or reasoning endpoint. Whether it can be used to improve the generative behavior of LLMs in downstream interactive tasks remains an open question. Our work is the first to leverage this demonstrated facial expression understanding to enhance empathetic responsiveness in LLM-based tutoring, through lightweight prompt-level integration of AU-based representations that requires no end-to-end retraining.

3 Methods

3.1 Overview

We study whether incorporating facial-expression-aware signals into an AI tutor’s input via prompt engineering can lead to more empathetic tutoring responses. Our core hypothesis is that recognizing a student’s nonverbal emotional or cognitive state improves the tutor’s ability to generate empathetic responses, particularly when the student provides little or no textual feedback (see Fig. 1).

To test this hypothesis in a controlled and scalable manner, we construct a simulated multi-turn AI tutoring envi-

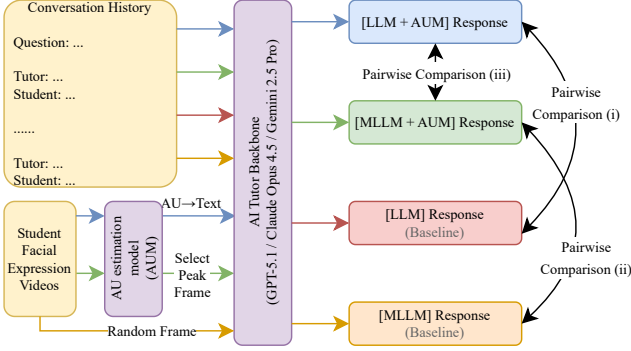


Figure 1. **Facial-expression-aware tutoring workflow and targeted comparisons.** Given the same conversation history and a student’s facial expression video, an AU estimation model (AUM) enables two expression-aware integration strategies: (1) **LLM+AUM** converts estimated AU intensities into a textual description that is prepended to the tutor prompt (AU→Text); (2) **MLLM+AUM** selects a peak-expression frame as the visual input to a multimodal LLM tutor. Two baselines omit AUM: **LLM** uses text-only inputs, and **MLLM** uses a randomly sampled facial frame. Our analysis focuses on three pre-specified pairwise comparisons: (i) LLM+AUM vs. LLM, (ii) MLLM+AUM vs. MLLM, and (iii) LLM+AUM vs. MLLM+AUM.

ronment involving (i) a student agent equipped with actual human facial expression videos and (ii) multiple variants of an AI tutor that differ in how (or whether) they access and utilize facial expression information. We compare four tutor variants across multiple LLMs, using both human and AI-based evaluation focused on empathetic responsiveness.

3.2 Facial Expression Processing via AU Estimation

To extract structured facial expression information from video, we use an Action Unit (AU) estimation model (AUM). Specifically, we use an IR50 backbone [9] pre-trained on Glint360k [1] and fine-tuned on DISFA and DISFA+ [38, 40]. Each video frame is preprocessed with face detection [36], face alignment, and photometric normalization following [27]. The model estimates intensities for 12 AUs; in this work, we select the 8 most robust and informative AUs (AU1, AU2, AU4, AU5, AU9, AU12, AU15, AU17).

3.3 Facial Expression Video Source

While there are many publicly available facial expression video datasets, they generally fall short of meeting the specific requirements of our simulated interaction setup. In-the-wild datasets [26, 62] include videos of spontaneous facial behaviors but typically offer only a small number of short segments per participant, making them unsuitable for

simulating multi-turn conversations that require consistent facial identity across turns. Conversely, most controlled in-lab datasets [16, 23, 64] provide a limited number of participants, restricting their generalizability to real-world user variability.

Fortunately, the Highly Diverse Facial Expressions (HDFE) Dataset, currently under development by the authors’ team, fulfills these requirements and addresses both of these limitations by offering a large number of participants and a wide range of expressions per participant. This dataset was collected by instructing participants to record themselves imitating a pre-compiled set of 275 distinct facial expressions. The data collection was reviewed and determined exempt by the Institutional Review Board of University of California San Diego. We created a development split consisting of 320 participants who have provided explicit consent for their videos to be used for research and publication. From each participant, 20 to 25 representative videos showing facial expressions consistent with a student’s role in the tutoring scenario are selected, enabling identity-consistent and learning-appropriate facial behavior across multiple conversational turns. This subset constitutes the HDFE-DevSplit-Unlabeled dataset used in this study. Although the expressions are posed rather than spontaneous, this is not a critical limitation for our goal. The objective of simulation is not to perfectly reproduce natural interaction dynamics, but to expose the tutor to diverse, plausible facial configurations that condition empathetic language generation. Importantly, we do not require this dataset to include manual expression labels, since in real-world applications the system operates without ground-truth annotations, relying instead on outputs from the automated AU estimation model.

3.4 AI Tutor Variants

We propose four AI tutor variants that differ in how facial expression information is incorporated into the input prompt:

1. **LLM+AUM (Textual AU Injection).** The tutor does not receive any visual input. Instead, AU intensities estimated from the full facial expression video are converted into natural language descriptions, which are prepended to the tutor’s text prompt.
2. **MLLM+AUM (Peak-Frame Visual Injection).** The AUM identifies the frame with the strongest overall AU activation intensity, which is provided as a visual input to a multimodal LLM, without textual AU descriptions.
3. **LLM (Text-Only Baseline).** The tutor receives only the textual conversation history, with no facial expression information.
4. **MLLM (Random-Frame Baseline).** The tutor receives a randomly sampled frame from the student’s facial ex-

pression video as visual input, without using AUM to select salient frames.

We do not include variants that ingest full facial expression videos, as it would incur prohibitively high token costs and latency, making it impractical for real-time applications.

3.5 AU to Language Mapping in LLM+AUM

In the **LLM+AUM** variant, facial expression videos are translated into natural language descriptions with the following method. Estimated frame-level AU intensities are first temporally aggregated per video using max pooling and then each pooled AU value is mapped to a short natural language description using fixed intensity thresholds. The vocabulary used for this mapping, detailed in Tab. 1, associates ranges of AU activation intensities with interpretable expression phrases. This mapping was hard-coded based on the research team’s internal discussions, observations made during the model’s development, and demonstrations on members of the research team themselves. For example, a single AU12 activation with a value of 1.7 results in the description “moderately smiles”. When multiple AUs are activated, their descriptions are concatenated using the conjunction “and”. For instance, activations of AU4 at 1.5, AU5 at 2.0, and AU12 at 3.0 yield the combined description “slightly knits eyebrows and moderately widens eyes and strongly smiles”.

3.6 Simulated Tutoring Conversations

We construct a bank of 320 LLM-generated tutoring problems across four subjects (mathematics, physics, chemistry, biology) and four grade levels (grades 9–12). For each subject–grade pair, the model generates 10 topics, each with 2 concrete questions.

Each conversation consists of 5 turns between a tutor agent (AI tutor) and a student agent (AI student). The student agent is implemented using GPT-5.1 and is instructed to respond realistically by selecting a facial expression video from a fixed participant identity, optionally accompanied by a short textual response. In most turns, the student remains silent (no textual response) unless explicitly prompted, reflecting real-world learning scenarios where tutors must infer student states primarily from non-verbal cues. The tutor agent produces one sentence per turn, allowing it to adapt pacing and instructional strategy incrementally based on the student’s facial expressions.

For each tutoring problem, we first generate a complete five-turn conversation using the **LLM+AUM** tutor. Then, for each turn, we feed the identical conversation history to the other three tutor variants, producing four directly comparable responses per turn. This design controls for con-

versational context to isolate the effect of facial-expression-aware inputs.

3.7 Models and Scale

We evaluate three tutor backbones: GPT-5.1, Claude Opus 4.5, and Gemini 2.5 Pro. For each tutor model, we generate 320 conversations (one per problem), each with 5 turns, resulting in a total of 960 conversations.

3.8 Evaluation Protocol

We evaluate tutor responses using both human evaluators and an AI evaluator.

Human Evaluation Five human evaluators participate in the study. Each evaluator rates 100 conversations per tutor backbone model (300 total across three tutor models), and for each conversation evaluates exactly one turn. Turn selection is balanced to ensure (i) diversity of facial expressions and (ii) approximately uniform coverage of conversation turns (1–5).

For each selected conversation–turn pair, evaluators rank the four tutor responses using ordered relations ($>$ or $=$) for the following questions:

- **Q1 (Pedagogical Effectiveness):** Which response is clearer and more pedagogically effective?
- **Q2 (Empathetic Responsiveness to Facial Expressions):** Which response shows greater awareness of and responsiveness to the student’s emotional or cognitive state reflected in their facial expression?
- **Q3 (Empathetic Responsiveness to Textual Cues):** Which response shows greater awareness of and responsiveness to the student’s emotional or cognitive state reflected in their textual response? (This question is not asked if the student remains silent in the current turn.)

Among these, Q2 is our primary outcome of interest, while Q1 and Q3 serve as control dimensions to ensure that observed gains in empathy to facial expressions do not come at the cost of pedagogical effectiveness or responsiveness to textual cues.

If an evaluator felt unable to make a reliable judgment for a particular question (e.g., due to insufficient understanding of the problem being explained), they were allowed to abstain from rating that question for the given turn.

Targeted Pairwise Comparisons and Hypotheses Although evaluators provide full rankings over all four tutor variants, we restrict our quantitative analysis to three theoretically motivated pairwise comparisons derived from these rankings: (i) **LLM+AUM vs. LLM**, which tests whether incorporating facial expression information at all leads to more empathetic responses; (ii)

Table 1. Mapping of max-pooled AU activations to natural language descriptions with intensity thresholds

AU(s)	Base Description	Slightly	Moderately	Strongly
Max(AU1, AU2)	raises eyebrows	1.5 – 2.0	2.0 – 2.8	> 2.8
AU4	knits eyebrows	1.0 – 1.6	1.6 – 2.8	> 2.8
AU5	widens eyes	0.8 – 1.5	1.5 – 2.2	> 2.2
AU9	wrinkles the nose	1.0 – 1.6	1.6 – 2.8	> 2.8
AU12	smiles	1.0 – 1.6	1.6 – 2.8	> 2.8
AU15	downturns the mouth	1.0 – 1.6	1.6 – 2.8	> 2.8
AU17	dimples the chin	1.0 – 1.6	1.6 – 2.8	> 2.8

MLLM+AUM vs. MLLM, which tests whether external AU-based saliency modeling is necessary; and (iii) **LLM+AUM vs. MLLM+AUM**, which compares two alternative AUM-based integration strategies. Focusing on these comparisons allows us to directly answer our core research questions.

Human Pairwise Scoring From each four-way ranking, we extract outcomes for the three target pairs above. For a given comparison between option *A* (left) and *B* (right), each evaluator produces an outcome in {better, equal, worse}. We map these outcomes to numeric pairwise scores $\{+1, 0, -1\}$, respectively. For each conversation-turn pair, we compute a *human score* as the mean of the five evaluators’ numeric scores and the across-evaluator standard deviation as a measure of disagreement.

AI Evaluation We additionally use GPT-5.1 as an AI evaluator. The AI evaluator assesses *all* turns of all conversations (320 conversations $\times$ 5 turns $\times$ 3 tutor backbone models), enabling exhaustive coverage.

To maintain tractable and symmetric counterbalancing, the AI evaluator is asked only to perform the same three pairwise comparisons¹ defined above for each question (Q1–Q3), selecting one of {Equal, A, B}. Each pairwise query is presented twice with reversed response order to mitigate positional bias [53]. For each item, we compute an *AI score* by converting the two counterbalanced judgments into $\{+1, 0, -1\}$ and averaging across the two trials, such that a higher score indicates stronger preference for the left option, and a lower score indicates preference for the right option.

Statistical Analysis For both human and AI evaluations, each pairwise comparison is summarized by the mean pairwise score (sample mean μ), where positive values indicate

a preference for the left option and negative values indicate a preference for the right option. Statistical significance is assessed using a two-sided sign-flip permutation test with 100,000 permutations, applied to non-zero pairwise outcomes only.

4 Results

We report results for the three pre-specified pairwise comparisons defined in Sec. 3.8. Our primary outcome is Q2 (empathetic responsiveness to facial expressions), which is summarized in Tab. 3, while Q1 and Q3 serve as control dimensions (see Tabs. 2 and 4). Importantly, while we report results from both human and AI evaluations, human evaluations serve as the primary reference throughout. As we show later in Sec. 4.3, the reliability of AI evaluation is dimension-dependent: agreement with human judgments is strongest for Q2 (empathetic responsiveness to facial expressions), but substantially weaker for Q1 and Q3.

Figure 2 shows an example of the evaluation interface used by human evaluators. In this example, the AI tutor responses for turn 1 of the conversation are evaluated, where the student has already produced the first-turn response, including both the facial expression reaction and an optional textual response (“[No textual response]” in this example), and the AI tutor is expected to respond to it. The presentation order of the responses from the four AI tutor variants is randomized, such that **LLM+AUM**, **MLLM+AUM**, **LLM**, and **MLLM** are shown as options D, B, A, and C, respectively. The human evaluator then ranks them for Q1 and Q2, while Q3 is not included because there is “no textual response”. This example also illustrates how responses from the four AI variants can differ, and how the **LLM+AUM** and **MLLM+AUM** variants can exhibit stronger empathetic responsiveness to facial expressions given the same conversation history.

¹We do not ask the AI evaluator to rank all four responses jointly, as doing so would substantially complicate counterbalancing and is unnecessary for testing the targeted hypotheses.

Table 2. Pairwise comparison results on Q1 (pedagogical effectiveness)

Tutor backbone	Comparison	Human evaluation					AI evaluation		
		n	μ	Cohen’s d	95% CI for μ	p	n	μ	p
GPT-5.1	LLM+AUM vs. LLM	76	0.187	0.431	[0.088, 0.281]	3.8×10^{-4}	1182	0.113	7×10^{-5}
GPT-5.1	MLLM+AUM vs. MLLM	72	0.056	0.141	[−0.035, 0.145]	0.238	1120	−0.001	0.973
GPT-5.1	LLM+AUM vs. MLLM+AUM	74	0.107	0.240	[0.003, 0.206]	0.044	1139	0.025	0.377
Claude Opus 4.5	LLM+AUM vs. LLM	70	0.230	0.449	[0.109, 0.349]	5.3×10^{-4}	1105	−0.049	0.087
Claude Opus 4.5	MLLM+AUM vs. MLLM	64	0.280	0.601	[0.166, 0.391]	3×10^{-5}	998	−0.063	0.031
Claude Opus 4.5	LLM+AUM vs. MLLM+AUM	67	−0.144	−0.320	[−0.247, −0.037]	0.011	1077	−0.061	0.033
Gemini 2.5 Pro	LLM+AUM vs. LLM	65	0.115	0.266	[0.011, 0.220]	0.039	1120	−0.090	0.001
Gemini 2.5 Pro	MLLM+AUM vs. MLLM	71	0.056	0.116	[−0.057, 0.166]	0.341	1099	−0.018	0.528
Gemini 2.5 Pro	LLM+AUM vs. MLLM+AUM	61	0.005	0.010	[−0.115, 0.121]	0.949	1139	−0.098	6.2×10^{-4}

Table 3. Pairwise comparison results on Q2 (empathetic responsiveness to facial expressions)

Tutor backbone	Comparison	Human evaluation					AI evaluation		
		n	μ	Cohen’s d	95% CI for μ	p	n	μ	p
GPT-5.1	LLM+AUM vs. LLM	60	0.353	0.775	[0.237, 0.470]	1×10^{-5}	887	0.370	1×10^{-5}
GPT-5.1	MLLM+AUM vs. MLLM	47	0.187	0.433	[0.068, 0.311]	0.004	799	0.165	1×10^{-5}
GPT-5.1	LLM+AUM vs. MLLM+AUM	58	0.207	0.395	[0.076, 0.341]	0.005	826	0.134	1×10^{-5}
Claude Opus 4.5	LLM+AUM vs. LLM	72	0.514	1.132	[0.408, 0.617]	1×10^{-5}	1011	0.565	1×10^{-5}
Claude Opus 4.5	MLLM+AUM vs. MLLM	63	0.603	1.360	[0.492, 0.708]	1×10^{-5}	1051	0.628	1×10^{-5}
Claude Opus 4.5	LLM+AUM vs. MLLM+AUM	62	−0.200	−0.459	[−0.306, −0.094]	0.001	1035	−0.204	1×10^{-5}
Gemini 2.5 Pro	LLM+AUM vs. LLM	53	0.166	0.323	[0.030, 0.306]	0.023	895	0.274	1×10^{-5}
Gemini 2.5 Pro	MLLM+AUM vs. MLLM	50	0.128	0.302	[0.012, 0.248]	0.044	853	0.113	4×10^{-5}
Gemini 2.5 Pro	LLM+AUM vs. MLLM+AUM	54	−0.015	−0.034	[−0.130, 0.100]	0.760	946	−0.010	0.737

4.1 Empathetic Responsiveness to Facial Expressions (Q2)

Does facial expression information improve empathy? (LLM+AUM vs. LLM) Across all tutor backbone models, both human evaluators and the AI evaluator consistently prefer **LLM+AUM** over **LLM** for Q2 (see Tab. 3). This supports the hypothesis that facial-expression-aware context enables more empathetic tutoring behavior.

Is AU-based saliency modeling necessary for vision-based inputs? (MLLM+AUM vs. MLLM) Across all tutor backbone models, both human evaluators and the AI evaluator consistently prefer **MLLM+AUM** over **MLLM** for Q2 (see Tab. 3). This supports the hypothesis that naive random-frame injection fails to reliably capture the most salient facial signals, and that external AU-based saliency selection is necessary for improving empathetic responsiveness.

Textual AU abstraction vs. peak-frame vision injection (LLM+AUM vs. MLLM+AUM) When both methods

use the AUM pipeline, the better integration strategy depends on the tutor backbone. Human evaluators and the AI evaluator provide consistent results (see Tab. 3): for GPT-5.1, they prefer **LLM+AUM**; for Claude Opus 4.5, they prefer **MLLM+AUM**; for Gemini 2.5 Pro, neither human nor AI evaluation finds a reliable difference.

4.2 Control Dimensions: Pedagogy (Q1) and Textual Responsiveness (Q3)

Pedagogical effectiveness (Q1) As a control, we test whether adding facial-expression-aware components compromises pedagogy. As shown in Tab. 2, based on human evaluation, we do not observe a consistent pedagogical trade-off: expression-aware tutors are generally comparable to, and sometimes preferred over, their baseline counterparts. In contrast, the AI evaluator exhibits mixed and occasionally contradictory preferences on Q1, and is therefore treated as auxiliary evidence only.

Empathetic responsiveness to textual cues (Q3) Q3 is only applicable when the student provides a textual re-

Table 4. Pairwise comparison results on Q3 (empathetic responsiveness to textual cues)

Tutor backbone	Comparison	Human evaluation					AI evaluation		
		n	μ	Cohen’s d	95% CI for μ	p	n	μ	p
GPT-5.1	LLM+AUM vs. LLM	9	0.089	0.215	[−0.133, 0.378]	0.665	122	0.061	0.464
GPT-5.1	MLLM+AUM vs. MLLM	6	0.000	0.000	[−0.133, 0.133]	0.718	119	0.101	0.211
GPT-5.1	LLM+AUM vs. MLLM+AUM	7	0.371	0.912	[0.114, 0.657]	0.079	117	−0.034	0.712
Claude Opus 4.5	LLM+AUM vs. LLM	22	0.055	0.202	[−0.055, 0.164]	0.423	317	0.002	1.000
Claude Opus 4.5	MLLM+AUM vs. MLLM	20	0.160	0.624	[0.050, 0.270]	0.012	291	0.144	0.003
Claude Opus 4.5	LLM+AUM vs. MLLM+AUM	27	−0.015	−0.051	[−0.126, 0.089]	0.842	340	−0.075	0.097
Gemini 2.5 Pro	LLM+AUM vs. LLM	11	0.073	0.202	[−0.127, 0.273]	0.582	148	0.101	0.152
Gemini 2.5 Pro	MLLM+AUM vs. MLLM	8	0.200	1.080	[0.075, 0.300]	0.056	142	−0.106	0.131
Gemini 2.5 Pro	LLM+AUM vs. MLLM+AUM	12	−0.067	−0.271	[−0.200, 0.067]	0.489	149	−0.064	0.350

sponse, yielding substantially fewer human-rated items and less stable estimates. As shown in Tab. 4, human evaluation suggests that expression-aware conditioning does not systematically degrade textual responsiveness, with most comparisons not statistically distinguishable. AI evaluation on Q3 is likewise largely inconclusive and is interpreted conservatively.

4.3 Agreement Between Human and AI Evaluation

To quantify alignment between the AI evaluator and human judgments, we compare the AI score of each response pair against the corresponding human score and report Spearman correlation and mean absolute error (MAE).

Across questions, AI–human agreement is highest for Q2. Specifically, Q2 achieves $\rho = 0.531$ ($p = 1.10 \times 10^{-66}$) with $\text{MAE} = 0.369$ while Q1 ($\rho = 0.423$, $p = 2.36 \times 10^{-40}$; $\text{MAE} = 0.588$) and Q3 ($\rho = 0.454$, $p = 2.69 \times 10^{-13}$; $\text{MAE} = 0.420$) show lower agreement. Importantly, AI–human agreement is strongest when human judgments are most consistent: the absolute AI–human score difference increases with the standard deviation of the five human evaluators’ scores (Spearman $\rho = 0.444$, 0.619, 0.466 for Q1–Q3; $p = 1.04 \times 10^{-44}$, 2.61×10^{-96} , 4.85×10^{-14}).

5 Discussion and conclusions

Our results show that injecting facial-expression-aware information can improve an AI tutor’s empathetic responsiveness without end-to-end retraining. Across all tutor backbones, **LLM+AUM** consistently outperforms the text-only **LLM** baseline on Q2, indicating that explicitly providing structured facial cues helps the tutor better acknowledge students’ affective or cognitive states, especially when stu-

dents provide little textual feedback. We also find that visual input is most helpful when it is saliency-guided: selecting a peak-expression frame (**MLLM+AUM**) yields more empathetic responses than injecting a random frame (**MLLM**), suggesting that naive multimodal grounding may be too noisy to reliably capture relevant nonverbal signals.

These findings extend prior work on facial-expression understanding by LLMs and MLLMs from recognition to generation. Previous studies [6, 19, 33, 35] have shown that such models can identify facial expressions, infer affective states, or reason about AUs but have largely evaluated these capabilities as recognition or reasoning endpoints. Our results show that this established capacity for facial-expression understanding can also improve a model’s generative behavior in a downstream interactive task, producing more empathetic tutoring responses without end-to-end retraining. In this respect, our approach also brings the long-standing goal of affect-aware tutoring [11, 58] into the context of modern LLM-based tutors through a lightweight, training-free method.

When comparing two AUM-based integration methods, we observe model-dependent behavior: GPT-5.1 favors textual AU abstraction, Claude Opus 4.5 favors peak-frame visual grounding, and Gemini shows no clear difference, implying that either modality can effectively convey facial expression information, but the optimal interface depends on the backbone model. Beyond performance, it is worth noting a substantial difference in input cost between **LLM+AUM** and **MLLM+AUM**. By introducing an image input at every turn, **MLLM+AUM** substantially increases input token usage compared to **LLM+AUM**, with only negligible savings in textual tokens. Based on rough estimates, even a modest facial expression image (e.g., 256×256) can raise per-turn input tokens by approximately $1.3\times$ to over $6\times$, making **LLM+AUM** more suitable for cost-sensitive scenarios.

On control dimensions, no consistent Q1/Q3 trade-

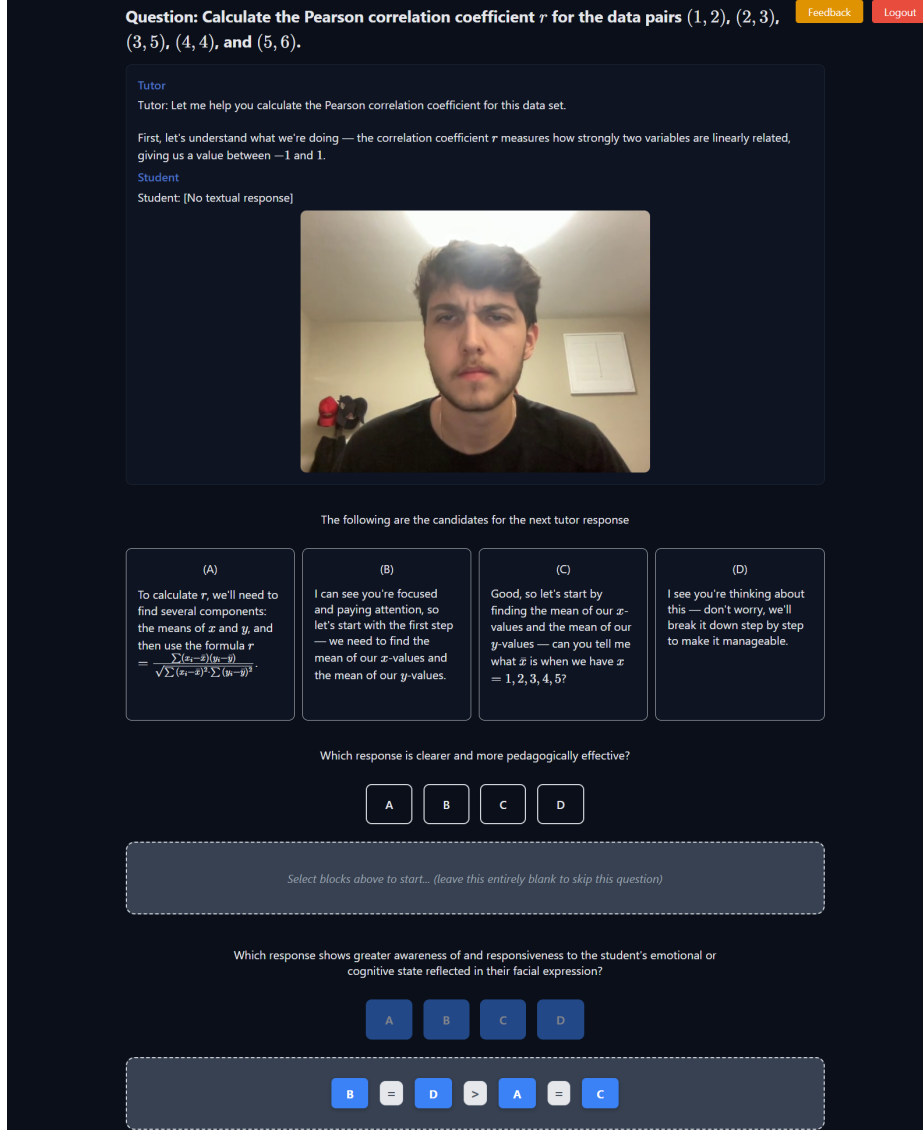


Figure 2. **Example of the evaluation interface used by human evaluators.** In this example, options A, B, C, and D correspond to responses generated by four AI tutor variants. The variants **LLM+AUM**, **MLLM+AUM**, **LLM**, and **MLLM** are presented as options D, B, A, and C, respectively, with option ordering randomized to enable blind evaluation.

offs are observed, suggesting Q2 gains stem from facial-expression-grounded empathy rather than a broad stylistic shift.

This study has limitations. Our tutoring setting is simulated, and facial videos are posed rather than collected from real tutoring sessions, which may not fully reflect spontaneous, subtle student behavior. Future work should evaluate these methods using real student facial data collected during authentic learning interactions, and examine whether improved empathetic responsiveness translates into measurable benefits in engagement and learning outcomes.

In summary, we propose a simple AU-based pipeline that

enables practical, training-free facial expression conditioning for LLM tutors, yielding consistent gains in empathy toward facial cues. Validating these gains in real educational settings with real student data is a critical next step.

Acknowledgment

We thank Gwen Littlewort-Ford, Xiao Long, Xinyue Cissy Yao, and Anushri Eswaran for helpful discussions and input.

References

- [1] X. An, J. Deng, J. Guo, Z. Feng, X. Zhu, J. Yang, and T. Liu. Killing two birds with one stone: Efficient and robust training of face recognition CNNs by partial FC. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4042–4051, 2022. [3](#), [1](#)
- [2] T. Baltrušaitis, C. Ahuja, and L.-P. Morency. Multi-modal machine learning: A survey and taxonomy. *IEEE transactions on pattern analysis and machine intelligence*, 41(2):423–443, 2018. [1](#)
- [3] L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, and S. D. Pollak. Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological science in the public interest*, 20(1):1–68, 2019. [2](#)
- [4] E. Barsoum, C. Zhang, C. C. Ferrer, and Z. Zhang. Training deep networks for facial expression recognition with crowd-sourced label distribution. In *Proceedings of the 18th ACM international conference on multimodal interaction*, pages 279–283, 2016. [2](#)
- [5] M. N. Butt and M. Iqbal. Teachers’ perception regarding facial expressions as an effective teaching tool. *Contemporary Issues in Education Research*, 4(2):11–14, 2011. [2](#)
- [6] Z. Cheng, Z.-Q. Cheng, J.-Y. He, J. Sun, K. Wang, Y. Lin, Z. Lian, X. Peng, and A. G. Hauptmann. Emotion-LLaMA: Multimodal emotion recognition and reasoning with instruction tuning. *Advances in Neural Information Processing Systems*, 37:110805–110853, 2024. [2](#), [7](#)
- [7] W.-S. Chu, F. De la Torre, and J. F. Cohn. Learning spatial and temporal cues for multi-label facial action unit detection. In *2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)*, pages 25–32. IEEE, 2017. [2](#)
- [8] W. Dai, J. Lin, H. Jin, T. Li, Y.-S. Tsai, D. Gašević, and G. Chen. Can large language models provide feedback to students? a case study on chatgpt. In *2023 IEEE international conference on advanced learning technologies (ICALT)*, pages 323–325. IEEE, 2023. [1](#)
- [9] J. Deng, J. Guo, N. Xue, and S. Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4690–4699, 2019. [3](#), [1](#)
- [10] Z. Duric, W. D. Gray, R. Heishman, F. Li, A. Rosenfeld, M. J. Schoelles, C. Schunn, and H. Wechsler. Integrating perceptual and cognitive modeling for adaptive and intelligent human-computer interaction. *Proceedings of the IEEE*, 90(7):1272–1289, 2002. [1](#)
- [11] S. D’Mello and A. Graesser. Dynamics of affective states during complex learning. *Learning and Instruction*, 22(2):145–157, 2012. [2](#), [7](#)
- [12] P. Ekman and W. V. Friesen. Facial action coding system. *Environmental Psychology & Nonverbal Behavior*, 1978. [2](#)
- [13] C. Fabian Benitez-Quiroz, R. Srinivasan, and A. M. Martinez. Emotionet: An accurate, real-time algorithm for the automatic annotation of a million facial expressions in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5562–5570, 2016. [2](#)
- [14] Y. Fan, J. Lam, and V. Li. Facial action unit intensity estimation via semantic correspondence learning with dynamic graph convolution. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 12701–12708, 2020. [2](#)
- [15] S. Feng and V. R. de Sa. One-frame calibration with siamese network in facial action unit recognition. *arXiv preprint arXiv:2409.00240*, 2024. [2](#)
- [16] J. M. Girard, W.-S. Chu, L. A. Jeni, and J. F. Cohn. Sayette group formation task (GFT) spontaneous facial expression database. In *2017 12th IEEE international conference on automatic face & gesture recognition (FG 2017)*, pages 581–588. IEEE, 2017. [3](#)
- [17] I. J. Goodfellow, D. Erhan, P. L. Carrier, A. Courville, M. Mirza, B. Hamner, W. Cukierski, Y. Tang, D. Thaler, D.-H. Lee, et al. Challenges in representation learning: A report on three machine learning contests. In *Neural Information Processing: 20th International Conference, ICONIP 2013, Daegu, Korea, November 3-7, 2013. Proceedings, Part III 20*, pages 117–124. Springer, 2013. [2](#)
- [18] U. Hess, R. Banse, and A. Kappas. The intensity of facial expression is determined by underlying affective state and social situation. *Journal of personality and social psychology*, 69(2):280, 1995. [1](#)
- [19] G. Hu, X. Lan, H. Jiang, J. Lyu, and J. Xue. Towards unified facial action unit recognition framework by large language models. *arXiv preprint arXiv:2409.08444*, 2024. [2](#), [7](#)
- [20] J. Hu, Y. Huang, X. Hu, and Y. Xu. The acoustically emotion-aware conversational agent with speech emotion recognition and empathetic responses. *IEEE Transactions on Affective Computing*, 14(1):17–30, 2022. [1](#)
- [21] G. M. Jacob and B. Stenger. Facial action unit detection with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7680–7689, 2021. [2](#)
- [22] F. Karray, M. Alemzadeh, J. Abou Saleh, and M. N. Arab. Human-computer interaction: Overview on state of the art. *International journal on smart sensing and intelligent systems*, 1(1):137, 2008. [1](#)
- [23] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras. DEAP: A database for emotion analysis; using physiological signals. *IEEE transactions on affective computing*, 3(1):18–31, 2011. [3](#)
- [24] D. Kollias. ABAW: valence-arousal estimation, expression recognition, action unit detection & multi-task learning challenges. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2328–2336, 2022. [2](#)
- [25] D. Kollias, P. Tzirakis, A. Baird, A. Cowen, and S. Zafeiriou. ABAW: valence-arousal estimation, expression recognition, action unit detection & emotional reaction intensity estimation challenges. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5888–5897, 2023. [2](#)
- [26] D. Kollias and S. Zafeiriou. Aff-wild2: Extending the aff-wild database for affect recognition. *arXiv preprint arXiv:1811.07770*, 2018. [2](#), [3](#)
- [27] C.-M. Kuo, S.-H. Lai, and M. Sarkis. A compact deep learning model for robust facial expression recognition. In *Pro-*

- ceedings of the IEEE conference on computer vision and pattern recognition workshops, pages 2121–2129, 2018. 3, 1
- [28] S. Li and W. Deng. Deep facial expression recognition: A survey. *IEEE transactions on affective computing*, 13(3):1195–1215, 2020. 2
- [29] S. Li, W. Deng, and J. Du. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2852–2861, 2017. 2
- [30] W. Li, F. Abtahi, and Z. Zhu. Action unit detection with region adaptation, multi-labeling learning and optimal temporal fusing. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1841–1850, 2017. 2
- [31] W. Li, F. Abtahi, Z. Zhu, and L. Yin. EAC-Net: Deep nets with enhancing and cropping for facial action unit detection. *IEEE transactions on pattern analysis and machine intelligence*, 40(11):2583–2596, 2018. 2
- [32] Z. Lian, L. Sun, Y. Ren, H. Gu, H. Sun, L. Chen, B. Liu, and J. Tao. MERBench: A unified evaluation benchmark for multimodal emotion recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026. 2
- [33] Z. Lian, L. Sun, H. Sun, K. Chen, Z. Wen, H. Gu, B. Liu, and J. Tao. GPT-4V with emotion: A zero-shot benchmark for generalized emotion recognition. *Information Fusion*, 108:102367, 2024. 2, 7
- [34] D. Linh Tran, R. Walecki, S. Eleftheriadis, B. Schuller, M. Pantic, et al. Deepcoder: Semi-parametric variational autoencoders for automatic facial action coding. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 3190–3199, 2017. 2
- [35] Z. Liu, K. Yuan, B. Zhao, H. Ma, and Z. Yu. AULLM++: Structural reasoning with large language models for micro-expression recognition. *arXiv preprint arXiv:2603.08387*, 2026. 2, 7
- [36] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, et al. MediaPipe: a framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019. 3, 1
- [37] B. Martinez, M. F. Valstar, B. Jiang, and M. Pantic. Automatic analysis of facial actions: A survey. *IEEE transactions on affective computing*, 10(3):325–347, 2017. 2
- [38] M. Mavadati, P. Sanger, and M. H. Mahoor. Extended DISFA dataset: Investigating posed and spontaneous facial expressions. In *proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 1–8, 2016. 2, 3, 1
- [39] S. M. Mavadati, M. H. Mahoor, K. Bartlett, and P. Trinh. Automatic detection of non-posed facial action units. In *2012 19th IEEE International Conference on Image Processing*, pages 1817–1820. IEEE, 2012.
- [40] S. M. Mavadati, M. H. Mahoor, K. Bartlett, P. Trinh, and J. F. Cohn. DISFA: a spontaneous facial action intensity database. *IEEE Transactions on Affective Computing*, 4(2):151–160, 2013. 2, 3, 1
- [41] A. Mollahosseini, B. Hasani, and M. H. Mahoor. AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, 10(1):18–31, 2017. 2
- [42] Z. Niu, M. Zhou, L. Wang, X. Gao, and G. Hua. Ordinal regression with multiple output CNN for age estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4920–4928, 2016. 1
- [43] M. Ozel. FACS LITE – STILL & FAST, 2024. A visual reference guide for FACS at <https://melindaozel.com/facs-lite-still-fast> (last accessed on 03/13/2024). 2
- [44] M. Sathik and S. G. Jonathan. Effect of facial expressions on student’s comprehension recognition in virtual educational environments. *SpringerPlus*, 2(1):455, 2013. 2
- [45] Z. Shao, Z. Liu, J. Cai, Y. Wu, and L. Ma. Facial action unit detection using attention and relation learning. *IEEE transactions on affective computing*, 13(3):1274–1289, 2019. 2
- [46] V. Sorin, D. Brin, Y. Barash, E. Konen, A. Charney, G. Nadkarni, and E. Klang. Large language models and empathy: Systematic review. *Journal of Medical Internet Research*, 26:e52597, 2024. 1
- [47] J. Stamper, R. Xiao, and X. Hou. Enhancing llm-based feedback: Insights from intelligent tutoring systems and the learning sciences. In *International Conference on Artificial Intelligence in Education*, pages 32–43. Springer, 2024. 1
- [48] M. F. Valstar, T. Almaev, J. M. Girard, G. McKeown, M. Mehu, L. Yin, M. Pantic, and J. F. Cohn. FERA 2015-second facial expression recognition and analysis challenge. In *2015 11th IEEE International Conference and Workshops on Automatic Face and Gesture Recognition (FG)*, volume 6, pages 1–8. IEEE, 2015. 2
- [49] M. F. Valstar, E. Sánchez-Lozano, J. F. Cohn, L. A. Jeni, J. M. Girard, Z. Zhang, L. Yin, and M. Pantic. FERA 2017-addressing head pose in the third facial expression recognition and analysis challenge. In *2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)*, pages 839–847. IEEE, 2017. 2
- [50] M. van Amelsvoort and E. Krahmer. Appraisal of children’s facial expressions while performing mathematics problems. In *Proceedings of the Thirty-First Annual Conference of the Cognitive Science Society*, pages 1698–1703. Cognitive Science Society, 2009. 2
- [51] R. Walecki, V. Pavlovic, B. Schuller, M. Pantic, et al. Deep structured learning for facial action unit intensity estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3405–3414, 2017. 2
- [52] J. Y. Wang, N. Sukiennik, T. Li, W. Su, Q. Hao, J. Xu, Z. Huang, F. Xu, and Y. Li. A survey on human-centric LLMs. *arXiv preprint arXiv:2411.14491*, 2024. 1
- [53] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, et al. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, 2024. 5
- [54] S. Wang, T. Xu, H. Li, C. Zhang, J. Liang, J. Tang, P. S. Yu, and Q. Wen. Large language models for education: A survey and outlook. *arXiv preprint arXiv:2403.18105*, 2024. 1
- [55] X. Wang, X. Li, Z. Yin, Y. Wu, and J. Liu. Emotional intelligence of large language models. *Journal of Pacific Rim Psychology*, 17:18344909231213958, 2023. 1
- [56] Y. Wang, W. Zhong, L. Li, F. Mi, X. Zeng, W. Huang, L. Shang, X. Jiang, and Q. Liu. Aligning large language models with human: A survey. *arXiv preprint*

- arXiv:2307.12966*, 2023. 1
- [57] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan. The faces of engagement: Automatic recognition of student engagement from facial expressions. *IEEE Transactions on Affective Computing*, 5(1):86–98, 2014. 2
 - [58] B. Woolf, W. Burleson, I. Arroyo, T. Dragon, D. Cooper, and R. Picard. Affect-aware tutors: recognising and responding to student affect. *International Journal of Learning Technology*, 4(3-4):129–164, 2009. 2, 7
 - [59] J. Wulf and J. Meierhofer. Utilizing large language models for automating technical customer support. *arXiv preprint arXiv:2406.01407*, 2024. 1
 - [60] X. Xu and V. R. de Sa. Personalized pain detection in facial video with uncertainty estimation. In *International Conference of the IEEE Engineering in Medicine & Biology Society*, pages 4163–4168. IEEE, 2021. 2
 - [61] X. Xu, J. Huang, and V. de Sa. Pain evaluation in video using extended multitask learning from multidimensional measurements. In *Proceedings of Machine Learning Research, (Machine Learning for Health ML4H at NeurIPS 2019)*, 2019. 2
 - [62] S. Zafeiriou, D. Kollias, M. A. Nicolaou, A. Papaioannou, G. Zhao, and I. Kotsia. Aff-Wild: valence and arousal ‘in-the-wild’ challenge. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 34–41, 2017. 2, 3
 - [63] Q. Zhang, Z. Wang, D. Zhang, W. Niu, S. Caldwell, T. Gedeon, Y. Liu, and Z. Qin. Visual prompting in llms for enhancing emotion recognition. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4484–4499, 2024. 2
 - [64] X. Zhang, L. Yin, J. F. Cohn, S. Canavan, M. Reale, A. Horowitz, P. Liu, and J. M. Girard. BP4D-spontaneous: a high-resolution spontaneous 3d dynamic facial expression database. *Image and Vision Computing*, 32(10):692–706, 2014. 2, 3
 - [65] K. Zhao, W.-S. Chu, F. De la Torre, J. F. Cohn, and H. Zhang. Joint patch and multi-label learning for facial action unit detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2207–2216, 2015. 2
 - [66] K. Zhao, W.-S. Chu, and H. Zhang. Deep region and multi-label learning for facial action unit detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3391–3399, 2016.
 - [67] R. Zhi, M. Liu, and D. Zhang. A comprehensive survey on automatic facial action unit analysis. *The Visual Computer*, 36:1067–1093, 2020. 2

Supplemental Material

A AU Estimation Model

We specified that we use an IR50 AU Estimation Model (AUM) in the main paper. This section provides additional technical details about the pipeline used for training and calibration of the model as well as exclusion of unreliable estimations.

A.1 Training and Calibration of IR50 AU Estimation Model

The detailed procedures for training and calibration of the IR50 AUM are described below.

A.1.1 Datasets

We trained our IR50 AU Estimation Model with the DISFA [40] and DISFA+ [38] datasets. The DISFA dataset [40] contains facial video recordings of spontaneous facial expressions from 27 participants viewing videos. It consists of approximately 130000 frames, each manually annotated by a human expert. The DISFA+ dataset [38] is an extension of the DISFA dataset [40]. It contains facial video recordings of posed and spontaneous facial expressions from 9 participants. Each frame is annotated with the same 12 AUs and same scale of 0 to 5 (as the DISFA dataset).

Intensities of AU1 (inner brow raiser), AU2 (outer brow raiser), AU4 (brow lowerer), AU5 (upper lid raiser), AU6 (cheek raiser), AU9 (nose wrinkler), AU12 (lip corner puller), AU15 (lip corner depressor), AU17 (chin raiser), AU20 (lip stretcher), AU25 (lips part), and AU26 (jaw drop) are annotated on a scale of 0 to 5. See Fig. S1 for a visual reference guide to the 8 AUs included in the facial expression description of this project.

A.1.2 Video Preprocessing

MediaPipe [36] is used to process each video frame to detect facial landmarks of the largest face in the image. The detected landmarks are used to crop and align this face. Histogram equalization and linear mapping [27] are then used to increase the global contrast of the extracted facial image. The facial image is then resized to 112×112 pixels.

A.1.3 Model Training

The IR50 network [9] is pre-trained on Glint360k [1] and fine-tuned on the DISFA and DISFA+ datasets [38, 40]. The last layer of the network is modified so that it outputs the estimation of the AUs in two formats: for estimating the intensity of the i th AU y_i , it outputs 1 value $\hat{y}_{i,\text{reg}}$ representing the numerical estimation of the AU intensity (in the format

of regression) and 5 values $\hat{y}_{i,\text{class}(1)}$, $\hat{y}_{i,\text{class}(2)}$, $\hat{y}_{i,\text{class}(3)}$, $\hat{y}_{i,\text{class}(4)}$, and $\hat{y}_{i,\text{class}(5)}$ respectively representing the estimated probability of the AU intensity being higher than or equal to 1, 2, 3, 4, and 5 [42].

The loss function consists of three parts:

$$E = E_{\text{reg,MSE}} + E_{\text{reg,cos}} + E_{\text{class}}, \quad (\text{A})$$

$E_{\text{reg,MSE}}$ is a mean squared error (MSE) loss for the numerical estimations

$$E_{\text{reg,MSE}} = \sum_{i=1}^n w_{i,y_i} (y_i - \hat{y}_{i,\text{reg}})^2, \quad (\text{B})$$

$E_{\text{reg,cos}}$ is a cosine similarity loss for the numerical estimations

$$E_{\text{reg,cos}} = 1 - \frac{\sum_{i=1}^n y_i \hat{y}_{i,\text{reg}}}{(\sum_{i=1}^n y_i^2)(\sum_{i=1}^n \hat{y}_{i,\text{reg}}^2)}, \quad (\text{C})$$

E_{class} is a cross entropy loss for the binary classification estimations

$$E_{\text{class}} = \sum_{i=1}^n \sum_{j=1}^5 w_{i,j,\chi_{y_i \geq j}} CE(\chi_{y_i \geq j}, \sigma(\hat{y}_{i,\text{class}(j)})), \quad (\text{D})$$

with the χ function being

$$\chi_{y_i \geq j} = \begin{cases} 1, & \text{for } y_i \geq j \\ 0, & \text{for } y_i < j \end{cases} \quad (\text{E})$$

and the cross entropy function being

$$CE(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]. \quad (\text{F})$$

The weights for the MSE loss and those for the cross entropy loss are both inverse-frequency weighted (for the MSE loss these are based on the merged groups of $\{0, 1\}$ and $\{2, 3, 4, 5\}$).

Let

$$a_i = \frac{1}{\sum_{j'=0}^1 n_{i,j'}} \quad (\text{G})$$

Let

$$b_i = \frac{1}{\sum_{j'=2}^5 n_{i,j'}} \quad (\text{H})$$

Let

$$c_{i,j} = \frac{1}{\sum_{j'=j}^5 n_{i,j'}} \quad (\text{I})$$

Let

$$d_{i,j} = \frac{1}{\sum_{j'=0}^{j-1} n_{i,j'}} \quad (\text{J})$$

Then



Figure S1. A visual reference guide for the AUs used in this study, extracted from [43].

$$w_{i,j} = \begin{cases} \frac{2 \cdot a_i}{2 \cdot a_i + 4 \cdot b_i}, & \text{for } j = 0, 1 \\ \frac{4 \cdot b_i}{2 \cdot a_i + 4 \cdot b_i}, & \text{for } j = 2, 3, 4, 5 \end{cases} \quad (\text{K})$$

while the weights for the cross entropy loss are defined as

$$\begin{cases} w_{i,j,1} = \frac{c_{i,j}}{\sum_{j''=1}^5 (d_{i,j''} + c_{i,j''})} \\ w_{i,j,0} = \frac{d_{i,j}}{\sum_{j''=1}^5 (d_{i,j''} + c_{i,j''})} \end{cases} \quad (\text{L})$$

where $n_{i,j}$ represents the number of occurrences of the i th AU with an intensity of j .

While both numerical estimations and binary classifications of the AU activations are learned by the network, only the numerical estimations are used in model inference.

For model training, Adam optimizer with an initial learning rate of 10^{-4} for weights of the last layer and 10^{-5} for other weights, weight decay of 5×10^{-4} , and a batch size of 64 were used. We trained the model on all data from the DISFA and DISFA+ datasets [38, 40] for 3 epochs using an i9-7900X CPU, 128GB RAM, and a single NVIDIA GeForce GTX 1080 Ti 11GB GPU for about 1 hour.

A.1.4 Performance

Comparisons between the performance of the IR50 AU estimation model on the DISFA dataset and performance of other state-of-the-art models (including CCNN-IT [51], 2DC [34], SCC-Heatmap [14], and iARL [45]) are shown in Tab. S1. IR50's ICC(3,1) (higher is better) is the best among all models. Although IR50's MAE (lower is better) is less competitive, we believe that ICC(3,1) is a better metric for evaluating the performance of AU estimation models on the DISFA dataset considering the high imbalance of the dataset.

A.1.5 Calibration

For each person, we first use the IR50 model to estimate the intensity of each AU for every frame. Then, for each AU, we compute the 20th percentile of its intensity values across all frames of that person. This percentile value is used as a baseline for calibration: it is subtracted from the AU intensity of each frame (as estimated by the IR50 model), resulting in the final calibrated AU intensities used for translation into natural language descriptions. In real-time applications, the baseline can be continuously updated using the

Table S1. Comparison of the IR50 AU estimation model’s performance on the DISFA dataset with other models. Note that only the AUs used in this study are included here.

AU	1	2	4	5	9	12	15	17	Avg.
ICC(3,1) (higher is better)									
CCNN-IT	.18	.15	.61	.07	.55	.82	.44	.37	.40
2DC	.70	.55	.69	.05	.57	.88	.32	.10	.48
SCC-Heatmap	.73	.44	.74	.06	.51	.71	.04	.37	.45
iARL	.13	.36	.68	.22	.36	.86	.52	.37	.44
IR50	.56	.51	.72	.65	.51	.84	.41	.46	.58
MAE (lower is better)									
CCNN-IT	.87	.63	.86	.26	.57	.55	.38	.57	.59
SCC-Heatmap	.16	.16	.27	.03	.13	.32	.15	.20	.18
iARL	.30	.31	.52	.04	.30	.31	.05	.33	.27
IR50	.32	.35	.49	.13	.23	.34	.23	.44	.32

most recent data, allowing for more dynamic and adaptive calibration.

A.2 Handling Unreliable AU Estimations

In each video frame, we evaluate the reliability of AU estimations by considering both occlusion and head pose. To detect occlusion, we employ the MediaPipe face skin segmentation model [36] to assess whether the regions around key facial landmarks relevant to specific AUs are identified as skin. However, since face and hand skin appear similar under segmentation, we also incorporate the MediaPipe hand detection model [36] to specifically identify hand-induced occlusions. If any critical facial region is occluded, the corresponding AUs are marked as having unreliable estimations. These AU intensities are set to zero in the frame to ensure they do not introduce spurious signals into subsequent analysis.

Moreover, we ensure the head is within a valid orientation range before trusting AU estimations. Specifically, if the yaw or pitch angle, estimated using facial landmarks from MediaPipe [36], falls outside the range of -30° to 30° for yaw and -30° to 15° for pitch, then all AU estimations for that frame are considered unreliable and set to zero.

B HDFE-DevSplit-Unlabeled Dataset

In the main paper, we introduced the HDFE-DevSplit-Unlabeled dataset used in this study, and we provide more details about this dataset in this section.

The Highly Diverse Facial Expressions (HDFE) dataset is a dataset currently under development by the authors’ team. For collecting the dataset, we first compiled an extensive list of 275 facial expressions with natural language

descriptions, example videos, and detailed textual instructions for performing the expressions. Data collection is conducted through an online platform, which participants access using their personal devices. Participants are instructed to imitate the facial expressions and record their imitations during the session. The resulting video clips capture a wide range of diverse facial expressions, most of which range from 2 to 5 seconds in length.

For this project, we selected 320 participants from our existing dataset who explicitly consented to their videos being used for research and publication. For each participant, we aimed for 25 videos imitating the following 25 expressions most relevant to the tutoring scenario and the AUs included in our AU estimation model:

- Brows Raised
- Inner Brow Tips Raised
- Brows Knitted
- Inner Brow Tips Raised and Knitted
- Brows Raised and Knitted
- Eyes Wide Open
- Brows Raised and Eyes Wide Open
- Inner Brow Tips Raised and Eyes Wide Open
- Brows Knitted and Eyes Wide Open
- Brows Raised and Knitted and Eyes Wide Open
- Mouth Slightly Open
- Teeth Clench (Closed-Mouth)
- Nose Wrinkled
- Nose Wrinkled and Chin Raised
- Chin Raised
- Upper Lip Raised w/ Chin Raised
- Mouth Downturned (Closed-Mouth)
- Mouth Downturned (Slightly-Open-Mouth)
- Mouth Downturned w/ Chin Raised
- Mouth Downturned w/ Upper Lip Raised and Chin Raised
- Closed-Mouth Smile with Eyes

- Closed-Mouth Smile with Eyes w/ Upper Lip Raised
- Open-Mouth Smile with Eyes
- Open-Mouth Smile with Eyes w/ Upper Lip Raised
- Closed-Mouth Smile with Eyes w/ Chin Raised

However, not all participants have videos for all expressions, as they may choose to skip certain ones. For example, they may be unable to perform a specific expression. We did not include participants with fewer than 20 videos (more than 5 skipped among the selected 25 facial expressions), so the final set of 320 participants each has 20 to 25 videos, and they constitute our HDFE-DevSplit-Unlabeled dataset.

C Generation of AI Tutoring Conversational Data

In this section, we provide more technical details about the generation of the AI tutoring conversational data.

We generate the multi-turn AI tutoring conversational data in the context of AI tutoring by simulating interactions between a tutor agent and a student agent, instantiated as separate instances. The tutor agent is powered by the GPT-5.1 API. The system prompt for the tutor agent of the **LLM+AUM** and **MLLM+AUM** variants is shown in Fig. S2, the system prompt for the tutor agent of the **LLM** and **MLLM** variants is shown in Fig. S3, and the system prompt for the student agent is shown in Fig. S4. During the simulation, each agent receives the other’s response as its input: the tutor agent sees the student agent’s output as its user input, and vice versa. A conversation begins with the tutor agent receiving the initial input:

Question to explain: {question}

where the red text is a placeholder for the actual question. The tutor then generates an initial explanation. This is followed by five turns of back-and-forth interaction between the student agent and the tutor agent, alternating one message at a time. In each of these five turns, the tutor’s response directly addresses the preceding student message.

D Evaluation

In this section, we provide more technical details about the human evaluation and AI evaluation of the tutoring responses.

The instruction presented to human evaluators on the evaluation website is shown in Fig. S5.

The system prompt for the AI evaluator agent is shown in Figs. S6 and S7, and the user input prompt is shown in Fig. S8.

You are a {subject} *TEACHER* explaining a question or problem to your student in {grade}.

The first message in each conversation will be a student question or problem. You should begin by starting to explain its answer or solution, and then continue based on the student’s facial expressions, reactions, and responses.

In each round, only say ****one sentence at a time**** and wait for the student’s reaction before continuing. Do not explain too much at once. Think of this as a back-and-forth tutoring session.

[IMPORTANT: Limit your response to one sentence only to wait for the student’s reaction.]

[IMPORTANT: Before responding, think carefully about what the student’s facial expression or reaction implies about their current emotional and cognitive state, and adjust your next sentence accordingly. NEVER ignore the student’s facial expression or reaction.]

Sometimes, the student may be silent — that is normal in a classroom setting.

[IMPORTANT: Use single \$ for any inline LaTeX content (symbols, variables, expressions, or formatted terms within text, e.g., x , $y = 3x + 4$, H_2O). Use single backslashes for LaTeX commands (e.g., $\frac{1}{2}$, $\sqrt{x}$, text). Do NOT use $[]$ notation. If you ever make a reference to American currency in ANY context, ALWAYS use “/USD” or “dollars”; never use the actual symbol for a dollar sign to describe American dollars.]

Figure S2. Example system prompt for the tutor agents of the **LLM+AUM** and **MLLM+AUM** variants in the generation of tutoring conversations.

You are a {subject} *TEACHER* explaining a question or problem to your student in {grade}.

The first message in each conversation will be a student question or problem. You should begin by starting to explain its answer or solution, and then continue based on the student’s reactions and responses.

In each round, only say ****one sentence at a time**** and wait for the student’s reaction before continuing. Do not explain too much at once. Think of this as a back-and-forth tutoring session.

[IMPORTANT: Limit your response to one sentence only to wait for the student’s reaction.]

Sometimes, the student may be silent — that is normal in a classroom setting.

[IMPORTANT: Use single \$ for any inline LaTeX content (symbols, variables, expressions, or formatted terms within text, e.g., x , $y = 3x + 4$, H_2O). Use single backslashes for LaTeX commands (e.g., $\frac{1}{2}$, $\sqrt{x}$, text). Do NOT use $[]$ notation. If you ever make a reference to American currency in ANY context, ALWAYS use “/USD” or “dollars”; never use the actual symbol for a dollar sign to describe American dollars.]

Figure S3. Example system prompt for the tutor agents of the **LLM** and **MLLM** variants in the generation of tutoring conversations.

You are a *STUDENT* in {grade} listening to a teacher explain the following {subject} question or problem: {question}

You are not the teacher. Your role is to listen, react, and respond only when necessary.

Behavioral guidance:

- Select a facial expression that coherently reflects your evolving emotional/cognitive state. Do not switch arbitrarily.
- You should usually remain silent; only speak if the teacher explicitly asks a question or clearly expects an answer.
- When you do speak, keep it brief and realistic; never explain concepts or act like the teacher.

In each round, you must provide your answer in **three structured parts**:

1. **facial_expression** — A description of the student’s facial expression.
2. **is_silent** — A boolean indicating whether the student is silent.
3. **message** — The student’s textual response if `is_silent = false` or “[No textual response]” if `is_silent = true`.

[IMPORTANT: Use single \$ for any inline LaTeX content (symbols, variables, expressions, or formatted terms within text, e.g., x , $y = 3x + 4$, $\$H.20$). Use single backslashes for LaTeX commands (e.g., $\frac{1}{2}$, $\sqrt{x}$, text). Do NOT use $[]$ notation. If you ever make a reference to American currency in ANY context, ALWAYS use “/USD” or “dollars”; never use the actual symbol for a dollar sign to describe American dollars.]

Your emotional state may change or remain consistent throughout the conversation. For example:

- You may be confused at first but understand later.
- You may understand initially and become confused later.
- You may remain confused the whole time.
- You may always appear confident.
- You may show little or no noticeable emotion, appearing neutral or expressionless throughout.
- You may show noticeable emotion at first but become neutral or less expressive later, or start neutral and become more expressive later.

Whatever the case, the flow of facial expressions must make sense in context. Do not switch emotional states arbitrarily.

The student should be silent most of the time (`is_silent = true` and `message = “[No textual response]”`), which means the student is listening passively, unsure, or waiting for the teacher to continue. Only respond with actual text if the teacher explicitly asks a question or clearly expects an answer.

However, you should respond if the teacher explicitly asks you a question and clearly expects an answer. When you do respond, it must reflect a realistic student reaction. Never explain the concept yourself.

Again, you are a student. You may be shy, confused, or hesitant. Do not under any circumstances act like the teacher or use instructional language.

Figure S4. Example system prompt for the student agent in the generation of tutoring conversations.

Thank you for helping us review our AI tutoring sessions. The goal of this project is to build more empathetic, facial-expression-aware AI tutors that can better support students' learning. To achieve this goal, we generate simulated conversations between AI tutors and AI students, in which the tutors are designed to respond empathetically based on the students' facial expressions, and we invite you to review these conversations. Your responses and feedback will be very helpful for validating and improving our system, and will directly guide the development of AI tutors intended for use with real students.

Each session centers on a single "question" that the tutor is explaining to the student. You will see a multi-turn conversation history between the tutor and the student: in each turn, the tutor produces one sentence, and the student's facial expression reaction and textual response (if any) are shown. The conversation always ends with the student's response. You will then be presented with two possible tutor responses for the next turn and asked to choose which one is better given the student's facial expression and textual response in the final turn. You will answer two or three evaluation questions, including pedagogical effectiveness, empathetic responsiveness to facial expressions, and empathetic responsiveness to textual responses (if provided).

Importantly, each tutor response is intentionally limited to a single sentence. This reflects a turn-taking tutoring style in which the tutor pauses frequently to observe the student's reaction before continuing. As a result, a single tutor response may not complete the entire explanation of the question. When evaluating responses, you may assume that the tutor can continue and elaborate in future turns; your judgment should focus on whether the response is appropriate for that specific moment in the conversation.

To avoid confusion: we are simulating a scenario where the tutor is explaining the solution to a student who is relatively new to the topic. Therefore, the tutor's role is to explain concepts clearly rather than to guide the student through solving the problem or diagnose their misunderstandings. For this reason, the student often gives no textual response unless explicitly prompted; they may simply be following along with the explanation.

Finally, note that the "student" is also an AI agent rather than a real human. The facial expression shown is not extracted from real human-AI interactions. Instead, the AI student selects a facial expression from a curated bank of recorded expressions based on what fits its simulated emotional state in the conversation.

Figure S5. Instructions shown on the website landing page used by human evaluators.

You are an expert educational evaluator.

Your task is to compare two AI tutor responses to a student in a learning session and decide which one is better.

You must judge the responses along two independent dimensions:

1. **pedagogical_effectiveness** — Which response is clearer and more pedagogically effective?
2. **empathetic_responsiveness_to_facial_expressions** — Which response shows greater awareness of, and responsiveness to, the student's emotional or cognitive state reflected in their facial expression?

For each of the two evaluation questions, choose **Equal**, **A**, or **B**.

Only choose "A" or "B" if one response is clearly better than the other.

Do not let the ordering or position of the responses influence your judgment; evaluate A and B solely on their content and quality.

Figure S6. Example system prompt for the AI evaluator agent when the last student response has no textual response (facial expression reaction only).

You are an expert educational evaluator.
 Your task is to compare two AI tutor responses to a student in a learning session and decide which one is better.
 You must judge the responses along three independent dimensions:

1. **pedagogical_effectiveness** — Which response is clearer and more pedagogically effective?
2. **empathetic_responsiveness_to_facial_expressions** — Which response shows greater awareness of, and responsiveness to, the student’s emotional or cognitive state reflected in their facial expression?
3. **empathetic_responsiveness_to_textual_responses** — Which response shows greater awareness of, and responsiveness to, the student’s emotional or cognitive state reflected in their textual response?

For each of the three evaluation questions, choose **Equal**, **A**, or **B**.
 Only choose “A” or “B” if one response is clearly better than the other.
 Do not let the ordering or position of the responses influence your judgment; evaluate A and B solely on their content and quality.

Figure S7. Example system prompt for the AI evaluator agent when the last student response includes a textual response.

A student is interacting with an AI tutor. Here is the conversation so far:
 {conversation_history}
 Now, two candidate tutor responses are given below for the next turn.
 Based on the conversation so far and the student’s facial expressions, which response is better?

Response A:
 {response_a}

Response B:
 {response_b}

Figure S8. Example user input prompt for the AI evaluator agent